

A disproof of Conjecture 00000003730

Complete bipartite graphs do not maximize energy, and paths do not minimize it among trees

1 The conjecture and its reading

Conjecture 00000003730 reads: *The extremum of graph spectral energy: energy is the sum of absolute eigenvalues, with the maximum complete bipartite graphs and the minimal trees paths.*

The energy of a finite simple graph G is $E(G) = \sum_i |\lambda_i|$, the λ_i being the eigenvalues of its adjacency matrix (Wikipedia, “Graph energy”: “the energy of a graph is the sum of the absolute values of the eigenvalues of the adjacency matrix of the graph”). The statement is a conjunction:

- (i) for each n , the maximum energy is attained by a complete bipartite graph $K_{a,b}$, $a + b = n$.
We refute this over (i-a) all graphs on n vertices and (i-b) bipartite graphs on n vertices;
- (ii) for each n , among trees on n vertices the minimum energy is attained by the path P_n .

We do not treat other comparison classes for (i), for example graphs with a fixed number of edges.

2 Definitions (as in the Lean file)

For G a `SimpleGraph V` on a finite type, $A = G.\text{adjMatrix } \mathbb{R}$ is Hermitian (`isHermitian_adjMatrix`), and $E(G) = \sum_i |\lambda_i|$, where λ_i are Mathlib’s `IsHermitian.eigenvalues`, the eigenvalues with multiplicity given by the spectral theorem. K_n is `⊤` on `Fin n`. $K_{a,b}$ is `completeBipartiteGraph (Fin a) (Fin b)`. P_n is Mathlib’s `pathGraph n`. The star $K_{1,3}$ is the graph on `Fin 4` whose edges are the pairs $\{0, j\}$, $j \neq 0$. “Bipartite” is Mathlib’s `IsBipartite` (2-colourable), and “tree” is `IsTree`.

3 Spectral lemmas

Let $Av = \lambda v$ with $v \neq 0$ (Mathlib’s eigenvector basis). Then $\sum_i \lambda_i = \text{tr } A = 0$, and, by the spectral theorem $A = UDU^*$, $\sum_i \lambda_i^2 = \text{tr}(A^2) = \sum_v \deg v = 2|E|$.

Lemma 1 (K_n). *Every eigenvalue of K_n is $n - 1$ or -1 ; hence $E(K_n) = 2(n - 1)$ for $n \geq 1$.*

Proof. $(Av)_u = s - v_u$ with $s = \sum_w v_w$, so $(\lambda + 1)v_u = s$ for all u . Summing gives $(\lambda + 1)s = ns$. If $s \neq 0$ then $\lambda = n - 1$. If $s = 0$ then $\lambda = -1$, since $v \neq 0$. On $\{n - 1, -1\}$ we have $|\lambda| = ((n - 2)\lambda + 2(n - 1))/n$, so $E = ((n - 2) \text{tr } A + 2n(n - 1))/n = 2(n - 1)$. $\square$

Lemma 2 ($K_{a,b}$). *Every eigenvalue of $K_{a,b}$ satisfies $\lambda = 0$ or $\lambda^2 = ab$; hence $E(K_{a,b}) = 2\sqrt{ab} \leq a + b$.*

Proof. With $S = \sum_l v_{(l)}$ and $T = \sum_r v_{(r)}$ (left and right parts), $\lambda v_{(l)} = T$ and $\lambda v_{(r)} = S$, so $\lambda S = aT$ and $\lambda T = bS$, hence $(\lambda^2 - ab)S = (\lambda^2 - ab)T = 0$. If $\lambda \neq 0$ and $\lambda^2 \neq ab$, then $S = T = 0$ and $v = 0$. If $ab = 0$, all eigenvalues are 0. Otherwise $|\lambda| = \lambda^2/\sqrt{ab}$ on $\{0, \pm\sqrt{ab}\}$, so $E = \text{tr}(A^2)/\sqrt{ab} = 2ab/\sqrt{ab} = 2\sqrt{ab}$. Finally $2\sqrt{ab} \leq a + b$ by AM–GM. $\square$

Lemma 3 ($K_{1,3}$ and P_4). *Both are trees on 4 vertices. The eigenvalues of the star satisfy $\lambda = 0$ or $\lambda^2 = 3$, so $E(K_{1,3}) = 6/\sqrt{3} = 2\sqrt{3}$. The eigenvalues of P_4 satisfy $\lambda^4 - 3\lambda^2 + 1 = 0$, so $|\lambda| = (\lambda^2 + 1)/\sqrt{5}$ and $E(P_4) = (6 + 4)/\sqrt{5} = 2\sqrt{5}$.*

Proof. Each graph is connected and has $3 = 4 - 1$ edges (Mathlib `isTree_iff_connected_and_card`). For P_4 : $v_1 = \lambda v_0$, $v_0 + v_2 = \lambda v_1$, $v_1 + v_3 = \lambda v_2$, $v_2 = \lambda v_3$. A linear combination gives $(\lambda^4 - 3\lambda^2 + 1)v_0 = 0$, and $v_0 = 0$ forces $v = 0$. Then $(|\lambda|\sqrt{5})^2 = 5\lambda^2 = (\lambda^2 + 1)^2$, because $\lambda^4 = 3\lambda^2 - 1$. The star is analogous, and $\text{tr}(A^2) = 6$ for both. $\square$

4 Refutation

Theorem 4. (i-a) *For every $n \geq 3$ and $a + b = n$: $E(K_{a,b}) \leq n < 2(n - 1) = E(K_n)$. So no complete bipartite graph has maximum energy on $n \geq 3$ vertices.* (i-b) *P_4 is bipartite with $E(P_4) = 2\sqrt{5} > 4 \geq E(K_{a,b})$ for all $a + b = 4$.* (ii) *$K_{1,3}$ is a tree on 4 vertices with $E = 2\sqrt{3} < 2\sqrt{5} = E(P_4)$.*

5 Lean formalization

File `lean/Conjecture3730/Basic.lean` (386 lines, import Mathlib only, namespace C3730). Key results: `sum_eig_sq`, `trace_sq_adj`, `eig_top`, `energy_top` ($E(K_n) = 2(n - 1)$ for all $n \geq 1$), `eig_cb`, `energy_cb` ($E(K_{a,b}) = 2\sqrt{ab}$ for all a, b), `energy_cb_le`, `star4_isTree`, `pathGraph4_isTree`, `energy_star4`, `energy_P4`, `completeGraph_beats_completeBipartite` (all $n \geq 3$), and `main` : $\neg \text{MaxIsCompleteBipartite} \wedge \neg \text{MaxBipartiteIsCompleteBipartite} \wedge \neg \text{MinTreeIsPath}$. Here `MaxIsCompleteBipartite` says $\forall n \exists a, b, a + b = n \wedge \forall G \text{ on Fin } n, E(G) \leq E(K_{a,b})$; `MaxBipartiteIsCompleteBipartite` adds the hypothesis `G.IsBipartite`; and `MinTreeIsPath` says $\forall n \forall T \text{ on Fin } n, T.\text{IsTree} \rightarrow E(P_n) \leq E(T)$. `lake build` succeeds; `#print axioms main` gives `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

6 Relation to the earlier submission

PR #226 (orionsheep) argued that K_5 (energy 8) beats every complete bipartite graph on 5 vertices (at most $2\sqrt{6}$). The reviewer rejected it because “the Lean proves $4*5 = 5*4$, ‘True := trivial’, $8 = 4+4$, $64 > 24$; adjacency matrices, eigenvalues, energy, and complete bipartite graphs appear nowhere in Lean.” This submission defines energy from Mathlib’s adjacency matrix and Hermitian eigenvalues. It proves the energies of K_n and $K_{a,b}$ for all parameters, and of $K_{1,3}$ and P_4 . Part (i) is refuted for every $n \geq 3$, and part (ii), which #226 did not formalize, is refuted with Mathlib trees.